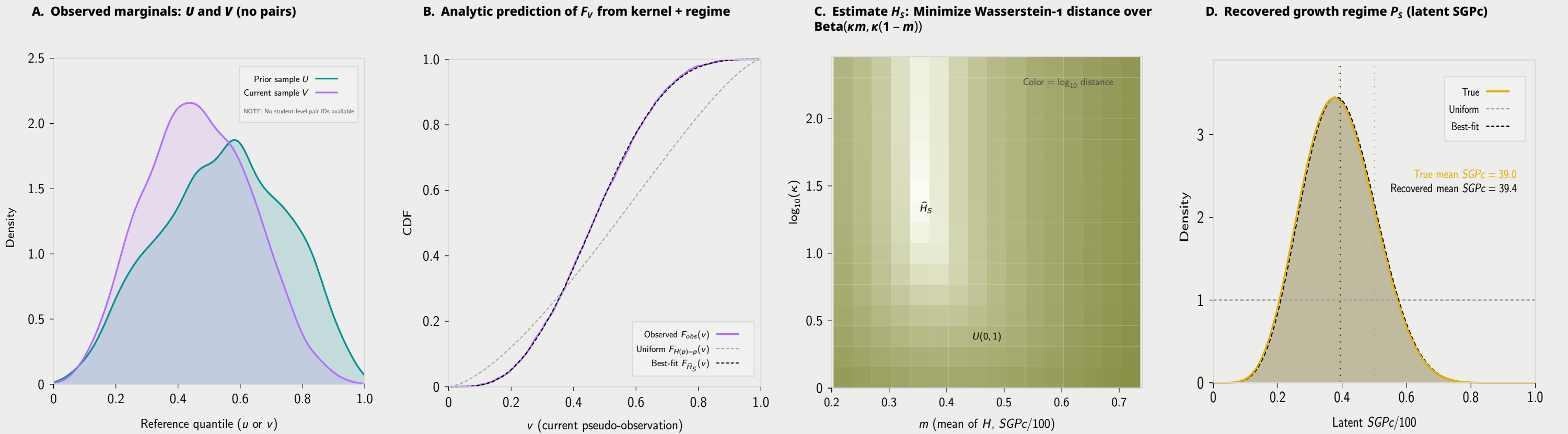


Growth as a stochastic process: From observed U and V to mean $SGPc$

Synthetic low-growth case (true mean $SGPc$ 39.0), baseline t -copula kernel ($\rho = 0.72$, $df = 8$)

$W_1(F_{V,S}^{obs}, F_H)$: Uniform = 0.0915, Best-fit = 0.0019 (97.9% reduction)



We observe two independent samples: prior- and current-scores, X and Y , respectively, for the same subgroup S (e.g., a TIMSS country or a NAEP state). We map scores to reference-quantiles using fixed reference marginals:

1. Prior reference quantiles $U = F_X^{\text{ref}}(x_i)$ and
2. Current reference quantiles $V = F_Y^{\text{ref}}(y_j)$.

Crucially, there is *no student-level linkage* between these samples—we cannot match any u_i to its corresponding v_j .

Panel A shows the two *univariate* marginals plotted on the same reference-quantile axis (“ u or v ”). This is the situation faced by TIMSS/NAEP-style designs when cohorts are sampled independently.

The distributions U and V are the only observables. All subsequent inference proceeds from these marginals plus an externally supplied copula dependence structure.

Let C_0 be a baseline copula (empirical or parametric) calibrated from longitudinal data. C_0 provides a *dependence template* for how students move from U to V and induces the conditional CDF (transition kernel)

$$F_0(v \mid u) = \Pr(V \leq v \mid U = u) = \frac{\partial}{\partial u} C_0(u, v).$$

Its inverse in v , the conditional quantile function/kernel, is

$$Q_0(p \mid u) = F_0^{-1}(p \mid u).$$

Intuitively, for each starting quantile u , the kernel $F_0(\cdot \mid u)$ describes the distribution of possible ending quantiles V under the baseline dependence structure C_0 .

Let $P \in [0, 1]$ be a random variable (i.e., a latent SGPc on the 0–1 scale) and p denote a realization in $[0, 1]$. A *growth regime*, H , is the distribution of P represented by its CDF:

$$H(p) := \Pr(P \leq p), \quad p \in [0, 1].$$

Given $U = u$ and a draw $P = p$, the model generates an outcome percentile by¹

$$V = Q_0(P \mid U = u).$$

A growth regime governs how the subgroup S “occupies” the baseline kernel $F_0(\cdot \mid u)$. The baseline growth regime is Uniform(0, 1) and its CDF is the identity on $[0, 1]$. That is,

$$P \sim \text{Uniform}(0, 1) \iff H(p) = p.$$

Departures from $H(p) = p$ tilt mass toward lower or higher conditional percentiles relative to the uniform baseline regime. Although we do not observe pairs (u_i, v_i) in TIMSS- or NAEP-style settings, the model provides a *forward map* from the prior to current margin.

Thus, for any fixed u ,

$$\begin{aligned} \Pr(V \leq v \mid U = u) &= \Pr(Q_0(P \mid u) \leq v) \\ &= \Pr(P \leq F_0(v \mid u)) \\ &= H(F_0(v \mid u)). \end{aligned}$$

Therefore, by the law of total probability, the subgroup’s predicted current-grade marginal CDF is

$$\begin{aligned} F_H(v) &= \Pr(V \leq v) \\ &= \mathbb{E}_U [H(F_0(v \mid U))] \\ &= \int_0^1 H(F_0(v \mid u)) dF_U(u) \\ &\approx \frac{\sum_{i=1}^n w_i H(F_0(v \mid u_i))}{\sum_{i=1}^n w_i}, \end{aligned}$$

where $\{u_i, w_i\}$ are the observed prior pseudo-observations and survey weights for the subgroup (i.e., a weighted empirical approximation to F_U).

We estimate the subgroup’s best-fit growth regime by searching over an admissible class $\mathcal{H}$ of candidate CDFs on $[0, 1]$. In this panel, $\mathcal{H}$ is the *Beta regime family*

$$\mathcal{H}_{\text{Beta}} = \{H_{m,\kappa}(\cdot) : P \sim \text{Beta}(\kappa m, \kappa(1 - m)), 0 < m < 1, \kappa > 0\},$$

where $m = \mathbb{E}[P]$ controls the regime center (mean SGPc on the 0–1 scale) and κ controls concentration (dispersion of latent SGPc).²

For each candidate $H \in \mathcal{H}$, Panel B provides the induced *predicted* current-grade marginal

$$F_H(v) = \mathbb{E}_U [H(F_0(v \mid U))].$$

The growth regime estimate for subgroup S is defined by the minimum-distance problem

$$\hat{H}_S = \underset{H \in \mathcal{H}_{\text{Beta}}}{\text{argmin}} W_1(F_{V,S}^{\text{obs}}, F_H),$$

where

$$W_1(F, G) = \int_0^1 |F(v) - G(v)| dv$$

is the Wasserstein-1 (earth mover’s) distance between the observed and predicted current-grade marginal CDFs.

The heat map displays $\log_{10}$ Wasserstein–1 distance over $(m, \log_{10} \kappa)$: lighter regions indicate candidate regimes whose induced F_H more closely matches $F_{V,S}^{\text{obs}}$. The marked optimum corresponds to $\hat{H}_S$ (equivalently, $(\hat{m}, \hat{\kappa})$). For reference, the uniform regime $H(p) = p$ occurs at $m = 0.5$ and $\kappa = 2$ (i.e., Beta(1, 1)), indicated by the $U(0, 1)$ marker.

The output of Panel C is a subgroup-specific *growth regime* H_S , where H_S is a cumulative distribution function (CDF) on $[0, 1]$ for a latent conditional-percentile random variable P_S (SGPc on the 0–1 scale):

$$H_S(p) = \Pr(P_S \leq p), \quad 0 \leq p \leq 1.$$

In words, H_S describes how subgroup S allocates conditional ranks *within* the baseline dependence template $F_0(\cdot \mid u)$.

Panel D shows the *distribution of P_S* (typically via its density $f_S(p) = H'_S(p)$ when H_S is smooth), not the CDF itself. Thus the “Best-fit” curve is the density corresponding to the fitted CDF H_S (i.e., $P_S \sim H_S$). The dashed grey “Uniform” reference corresponds to the uniform regime $H(p) = p$ (equivalently $P \sim \text{Unif}(0, 1)$), which places no tilt on conditional percentiles relative to the baseline kernel.

Because we restrict H to the Beta family in Panel C, $P_S \sim \text{Beta}(\kappa m, \kappa(1 - m))$: the mean m sets the regime center (reported on the SGPc scale as $100m$) and κ controls concentration (dispersion of latent SGPc).

The vertical reference lines mark the mean SGPc under the uniform regime (50), the generating “True” regime (), and the recovered best-fit regime (). In applications, the primary outputs are H_S (and uncertainty bands) and summary functionals such as mean/median SGPc and bucket membership probabilities.

¹ We assume $P \perp U$. That is, p is drawn independently of starting point within the subgroup. This assumption is relaxed in the full model.

² Other regime families considered include the truncated exponential family—the maximum entropy probability distribution for a random variable restricted to a finite interval with a fixed mean.